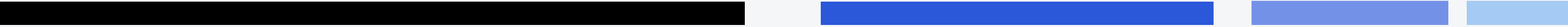


Lecture-4



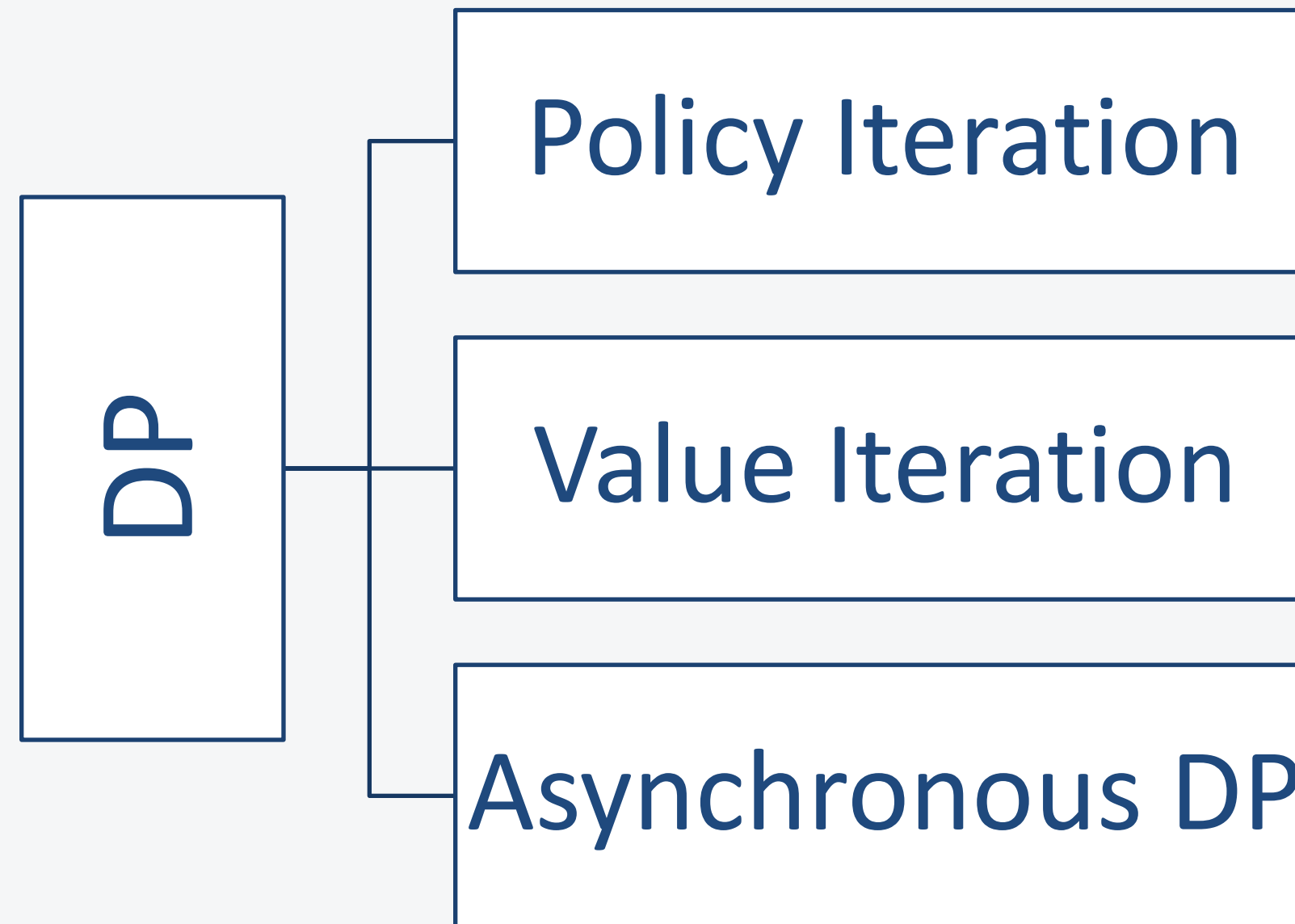
Model Free RL

Agenda

01. Dynamic Programming Review
02. Model Free RL
03. Monte Carlo Methods
04. Temporal Difference Method
05. On Policy Learning
06. Off Policy Learning
07. Example
08. Implementation

01.

Dynamic Programming (Review)



DP

Problem	Bellman Equation	Algorithm
Prediction	Bellman Expectation Equation	Iterative Policy Evaluation
Control	Bellman Expectation Equation + Greedy Policy Improvement	Policy Iteration
Control	Bellman Optimality Equation	Value Iteration

DP

Q: What conditions or assumptions are mandatory to apply DP?

Ans: 1. The environment is represented by Markov Decision Process.

Definition

A *Markov Decision Process* is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

- $\mathcal{S}$ is a finite set of states
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{P}$ is a state transition probability matrix,
 $\mathcal{P}_{ss'}^a = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a]$
- $\mathcal{R}$ is a reward function, $\mathcal{R}_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$
- γ is a discount factor $\gamma \in [0, 1]$.

DP

Q: What conditions or assumptions are mandatory to apply DP?

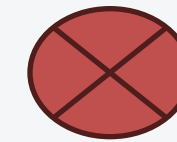
Ans: 2. The complete model of environment is available.

$$\mathcal{R}_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$$

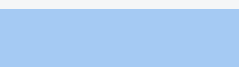
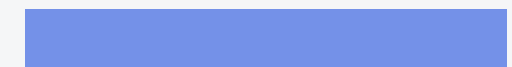
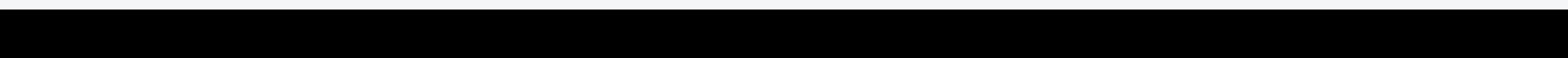
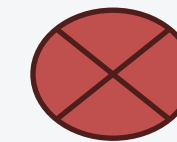
$$p(s', r \mid s, a) \doteq \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\},$$

Q: What if MDP condition is satisfied but we don't have the model?

$$\mathcal{R}_s^a = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$$



$$p(s', r \mid s, a) \doteq \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\},$$



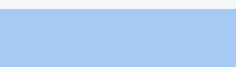
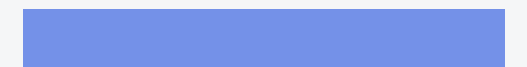
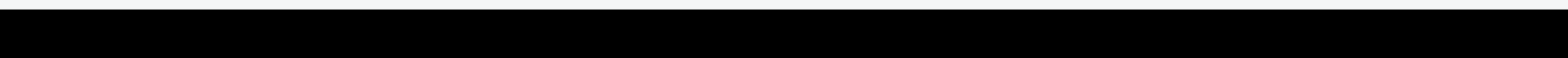
Bellman Equation

$$\begin{aligned} v_{\pi}(s) &\doteq \mathbb{E}_{\pi}[G_t \mid S_t = s] \\ &= \mathbb{E}_{\pi}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s'} \sum_r \underbrace{p(s', r | s, a)} \left[r + \gamma \mathbb{E}_{\pi}[G_{t+1} | S_{t+1} = s'] \right] \\ &= \sum_a \pi(a|s) \sum_{\underbrace{s', r}} p(s', r | s, a) \left[r + \gamma v_{\pi}(s') \right], \quad \text{for all } s \in \mathcal{S}, \end{aligned}$$

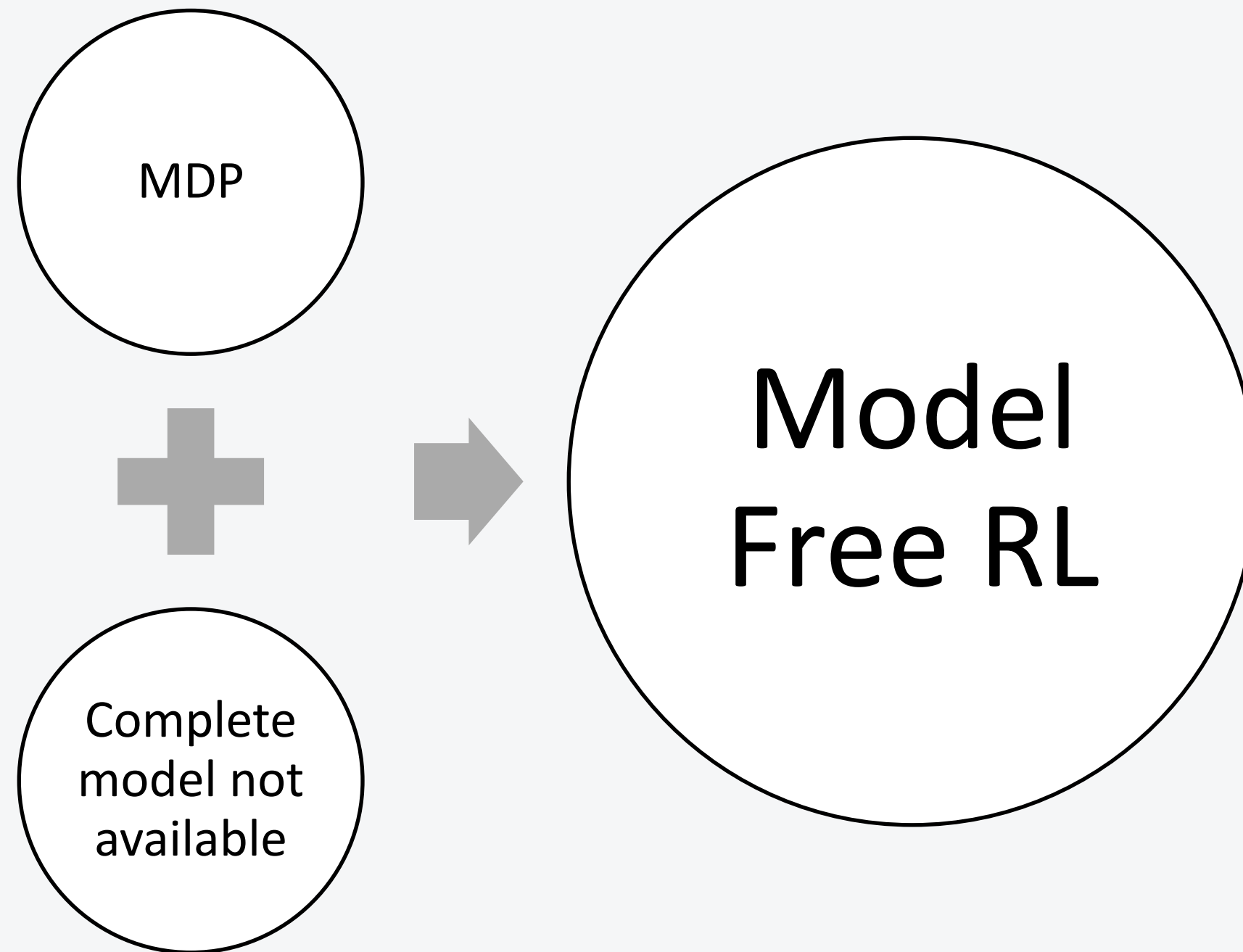
No more applicable

02.

Model Free RL

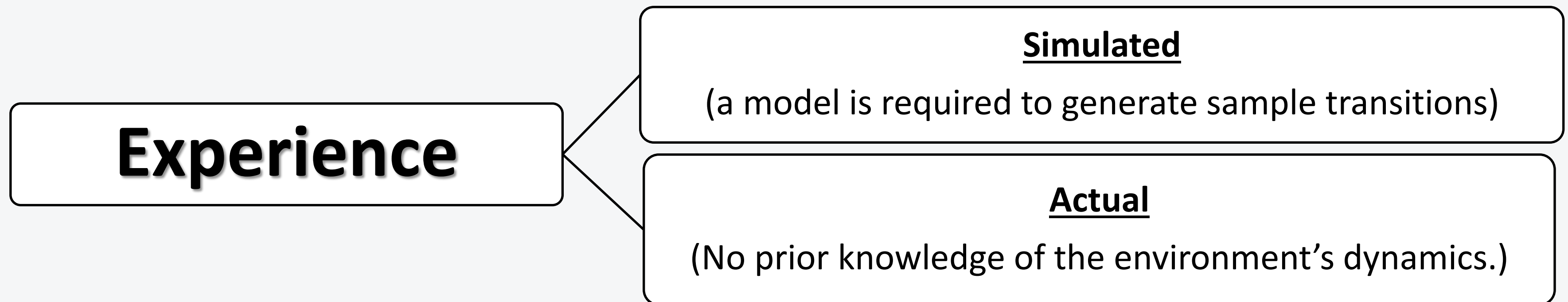


Model Free RL

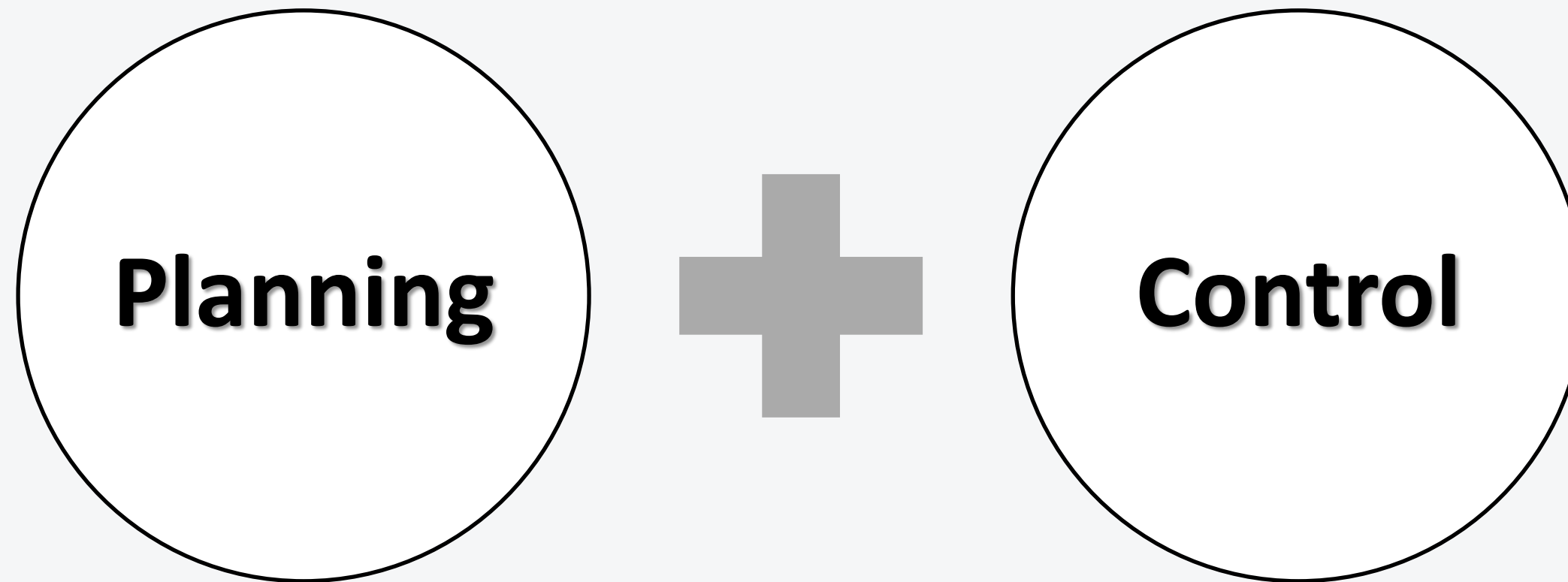


Model Free RL

- Experience—Sample sequences of states, actions, and rewards from actual or simulated interaction with an environment.

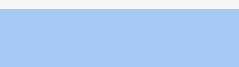
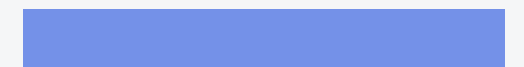
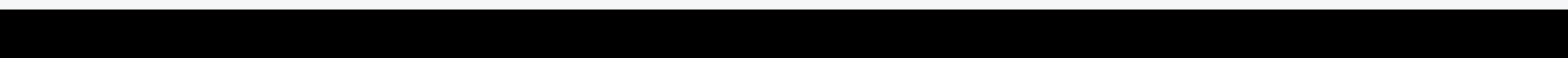


Model Free RL



03.

Monte Carlo Methods



Bellman Equation

$$\begin{aligned} v_{\pi}(s) &\doteq \mathbb{E}_{\pi}[G_t \mid S_t = s] \longleftarrow \text{How to get this now?} \\ &= \mathbb{E}_{\pi}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s'} \sum_r \underbrace{p(s', r \mid s, a)} \left[r + \gamma \mathbb{E}_{\pi}[G_{t+1} \mid S_{t+1} = s'] \right] \\ &= \sum_a \pi(a|s) \sum_{\underbrace{s', r}} p(s', r \mid s, a) \left[r + \gamma v_{\pi}(s') \right], \quad \text{for all } s \in \mathcal{S}, \end{aligned}$$

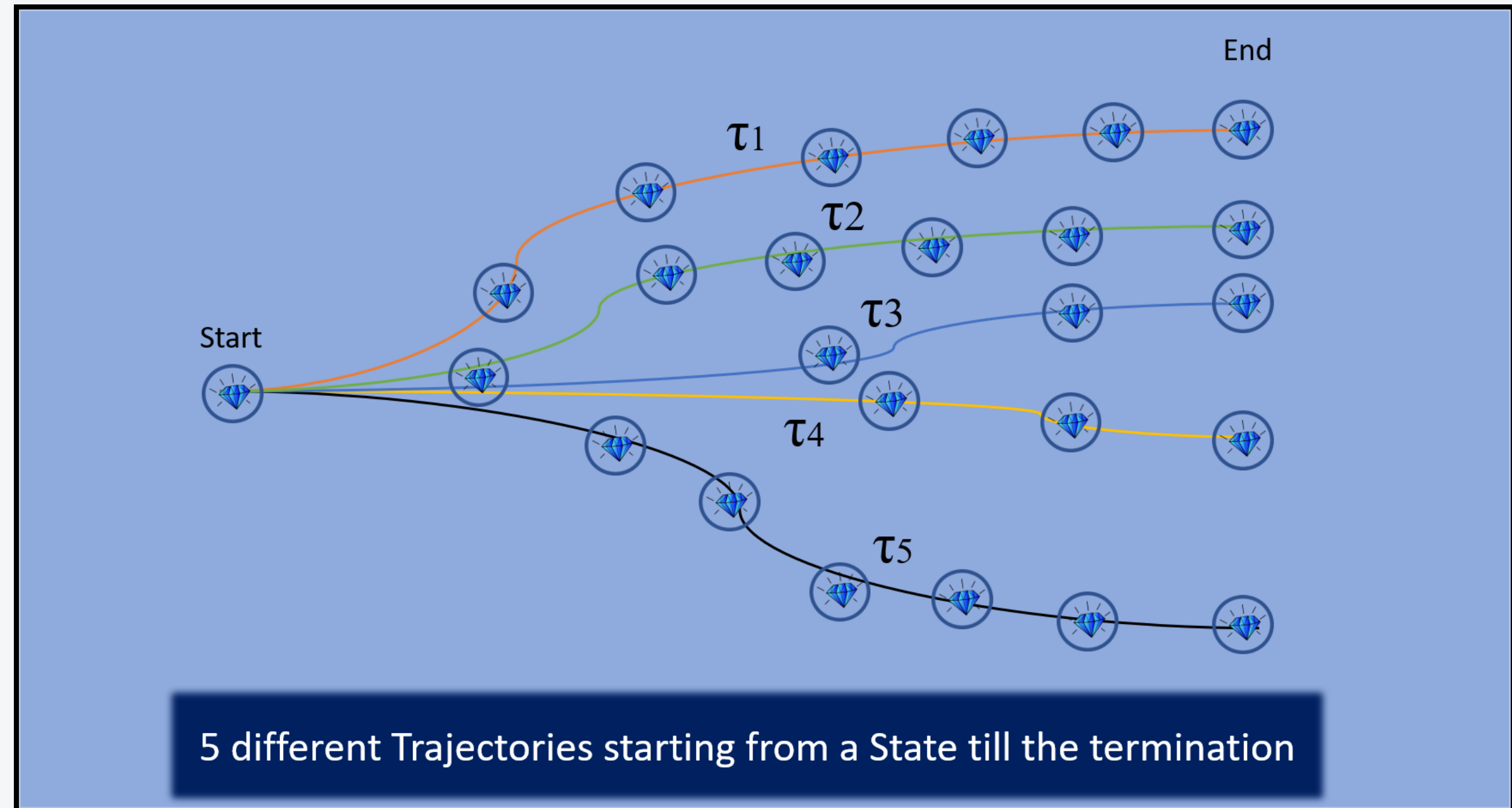
No more applicable

Law of Large Numbers

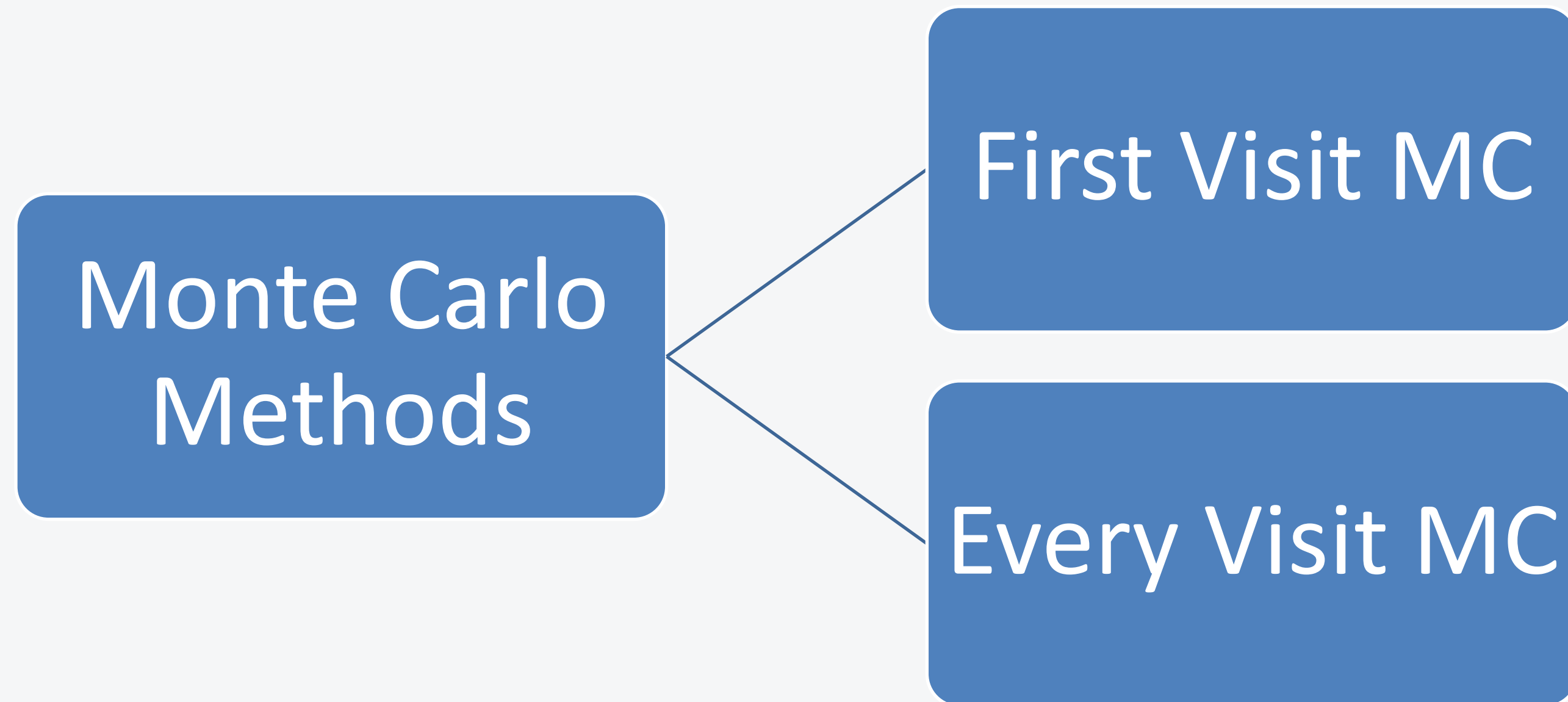
- The law of large numbers asserts that as a random experiment is repeated numerous times, the average of the outcomes will converge towards the expected value of those outcomes.
- This concept emphasizes the necessity of conducting **a large number of trials** to accurately observe the underlying probabilities of different outcomes.

Monte Carlo Methods

- Episodic problem.
- Complete episode.



Monte Carlo Methods



First Visit MC Prediction

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated

Initialize:

$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$

$Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless S_t appears in $S_0, S_1, \dots, S_{t-1}$:

Append G to $Returns(S_t)$

$V(S_t) \leftarrow \text{average}(Returns(S_t))$

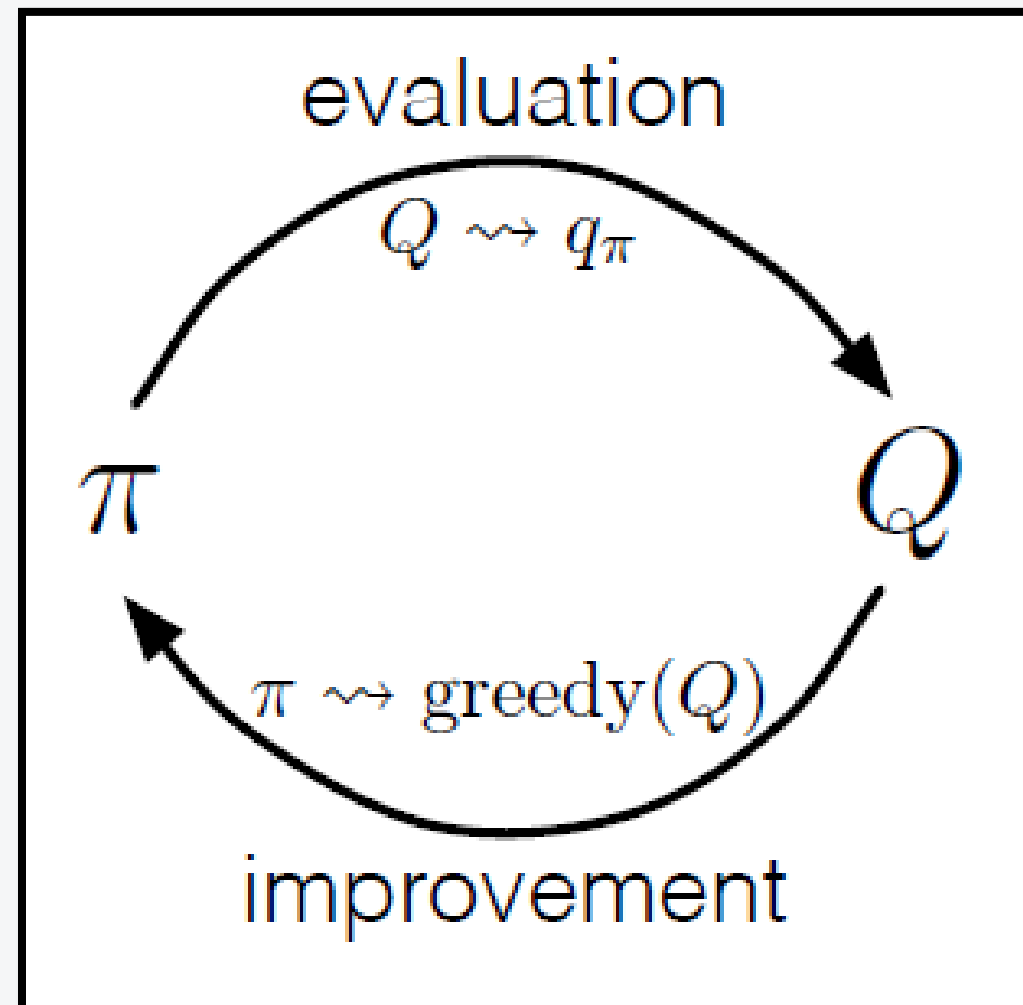
Incremental MC

After each episode $i = s_{i,1}, a_{i,1}, r_{i,1}, s_{i,2}, a_{i,2}, r_{i,2}, \dots$

- Define $G_{i,t} = r_{i,t} + \gamma r_{i,t+1} + \gamma^2 r_{i,t+2} + \dots$ as return from time step t onwards in i th episode
- For state s visited at time step t in episode i
 - Increment counter of total visits: $N(s) = N(s) + 1$
 - Update estimate

$$V^\pi(s) = V^\pi(s) \frac{N(s) - 1}{N(s)} + \frac{G_{i,t}}{N(s)} = V^\pi(s) + \frac{1}{N(s)} (G_{i,t} - V^\pi(s))$$

MC Control

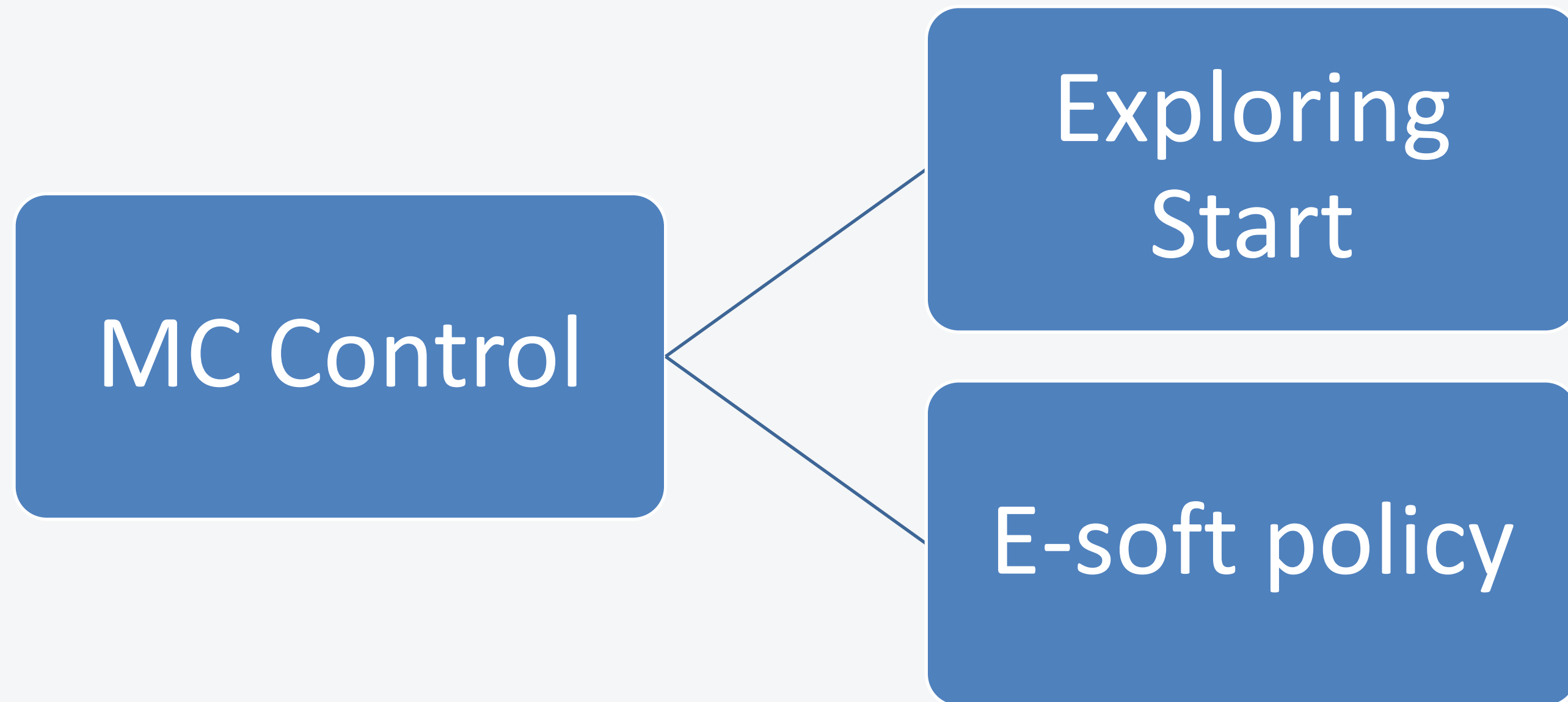


$$\pi_0 \xrightarrow{\text{E}} q_{\pi_0} \xrightarrow{\text{I}} \pi_1 \xrightarrow{\text{E}} q_{\pi_1} \xrightarrow{\text{I}} \pi_2 \xrightarrow{\text{E}} \dots \xrightarrow{\text{I}} \pi_* \xrightarrow{\text{E}} q_*,$$

MC Control

- A complication: many state–action pairs may never be visited.
- With no returns to average, the Monte Carlo estimates of the other actions will not improve with experience.
- This is a serious problem because the purpose of learning action values is to help in choosing among the actions available in each state.

MC Control



MC Control (Exploring Starts)

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:

$\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

Loop forever (for each episode):

Choose $S_0 \in \mathcal{S}, A_0 \in \mathcal{A}(S_0)$ randomly such that all pairs have probability > 0

Generate an episode from S_0, A_0 , following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$

MC Control (e-soft policy)

On-policy first-visit MC control (for ε -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ε -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

MC**First Visit****Every Visit****Bias****Unbiased****Asymptotically
Unbiased****Variance****Noisy****Low than FV****MSE**

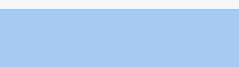
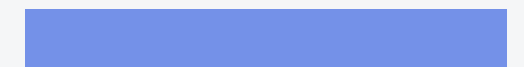
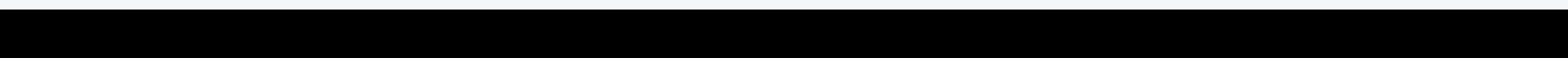
$$\text{MSE}(\hat{\theta}) = \text{Var}_{\theta}(\hat{\theta}) + \text{Bias}(\hat{\theta}, \theta)^2.$$

Do MC bootstrap?

- In Monte Carlo methods, the estimates for each state are independent.
- The estimate for one state does not build upon the estimate of any other state, as is the case in DP.
- **Hence, Monte Carlo methods do not bootstrap.**

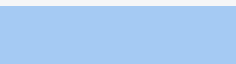
Computational Expense

- The computational expense of estimating the value of a single state is independent of the number of states.
- This can make Monte Carlo methods particularly attractive when one requires the value of only one or a subset of states.
- One can generate many sample episodes starting from the states of interest, averaging returns from only these states, ignoring all others.

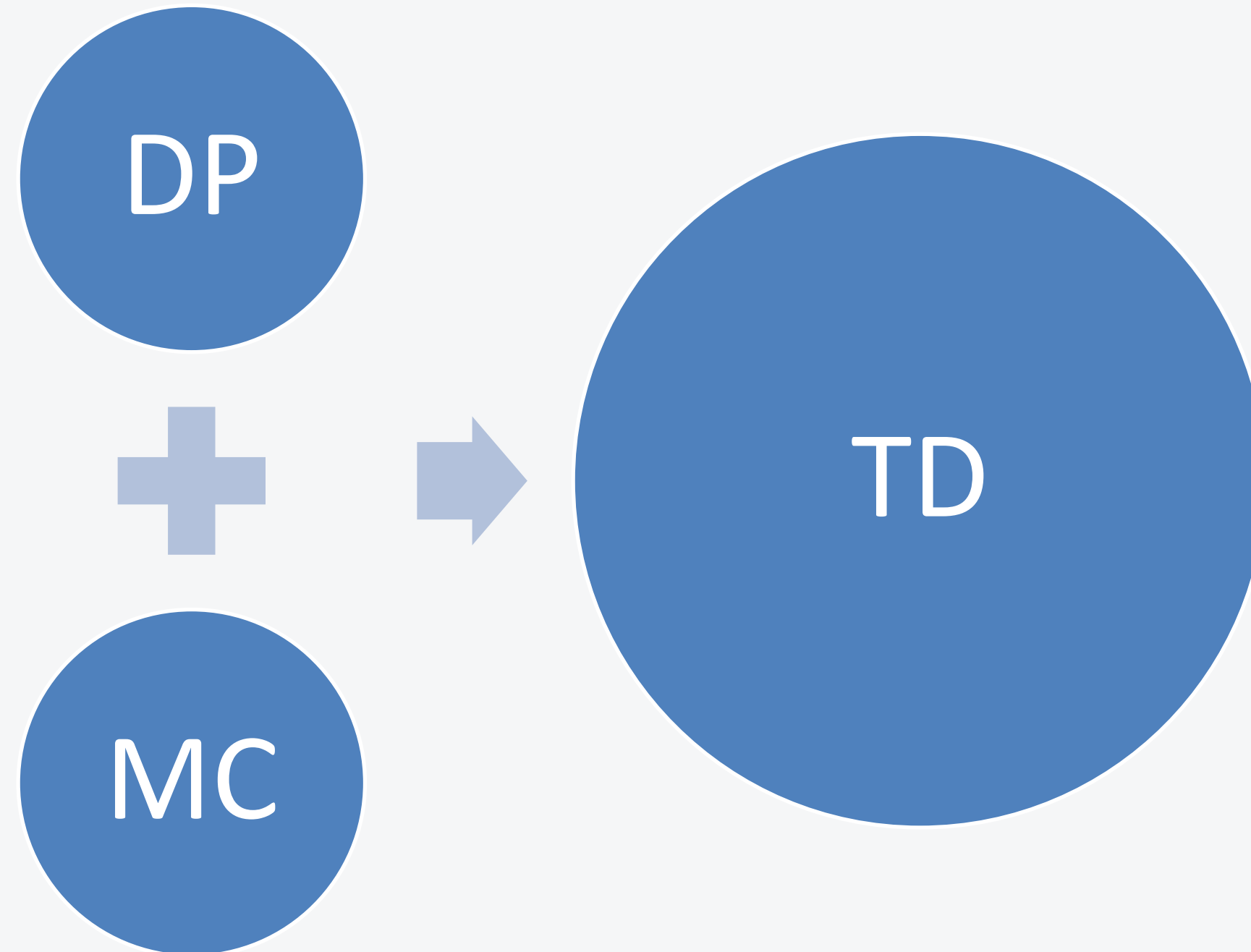


04.

Temporal Difference



Temporal Difference



Bellman Equation

$$\begin{aligned} v_{\pi}(s) &\doteq \mathbb{E}_{\pi}[G_t \mid S_t = s] \\ &= \mathbb{E}_{\pi}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \quad \leftarrow \text{How to get this now?} \\ &= \sum_a \pi(a|s) \sum_{s'} \sum_r \underbrace{p(s', r | s, a)} \left[r + \gamma \mathbb{E}_{\pi}[G_{t+1} | S_{t+1} = s'] \right] \\ &= \sum_a \pi(a|s) \sum_{\underbrace{s', r}} p(s', r | s, a) \left[r + \gamma v_{\pi}(s') \right], \quad \text{for all } s \in \mathcal{S}, \end{aligned}$$

No more applicable

Update Rules

MC

$$V(S_t) \leftarrow V(S_t) + \alpha [G_t - V(S_t)],$$

TD(0)

$$V(S_t) \leftarrow V(S_t) + \alpha [R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$

TD Algorithm

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated

Algorithm parameter: step size $\alpha \in (0, 1]$

Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$

Loop for each episode:

 Initialize S

 Loop for each step of episode:

$A \leftarrow$ action given by π for S

 Take action A , observe R, S'

$V(S) \leftarrow V(S) + \alpha[R + \gamma V(S') - V(S)]$

$S \leftarrow S'$

 until S is terminal

SARSA

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t) \right].$$

SARSA

Sarsa (on-policy TD control) for estimating $Q \approx q_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$

Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$

Loop for each episode:

 Initialize S

 Choose A from S using policy derived from Q (e.g., ε -greedy)

 Loop for each step of episode:

 Take action A , observe R, S'

 Choose A' from S' using policy derived from Q (e.g., ε -greedy)

$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma Q(S', A') - Q(S, A)]$

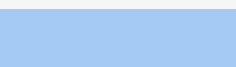
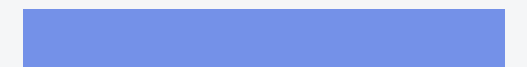
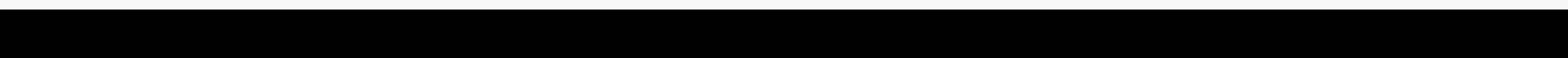
$S \leftarrow S'; A \leftarrow A';$

 until S is terminal

	First Visit	Every Visit	TD
Bias	Unbiased	Asymptotically Unbiased	Biased
Variance	Noisy	Low than FV	Lowest
MSE	$\text{MSE}(\hat{\theta}) = \text{Var}_{\theta}(\hat{\theta}) + \text{Bias}(\hat{\theta}, \theta)^2.$		

05.

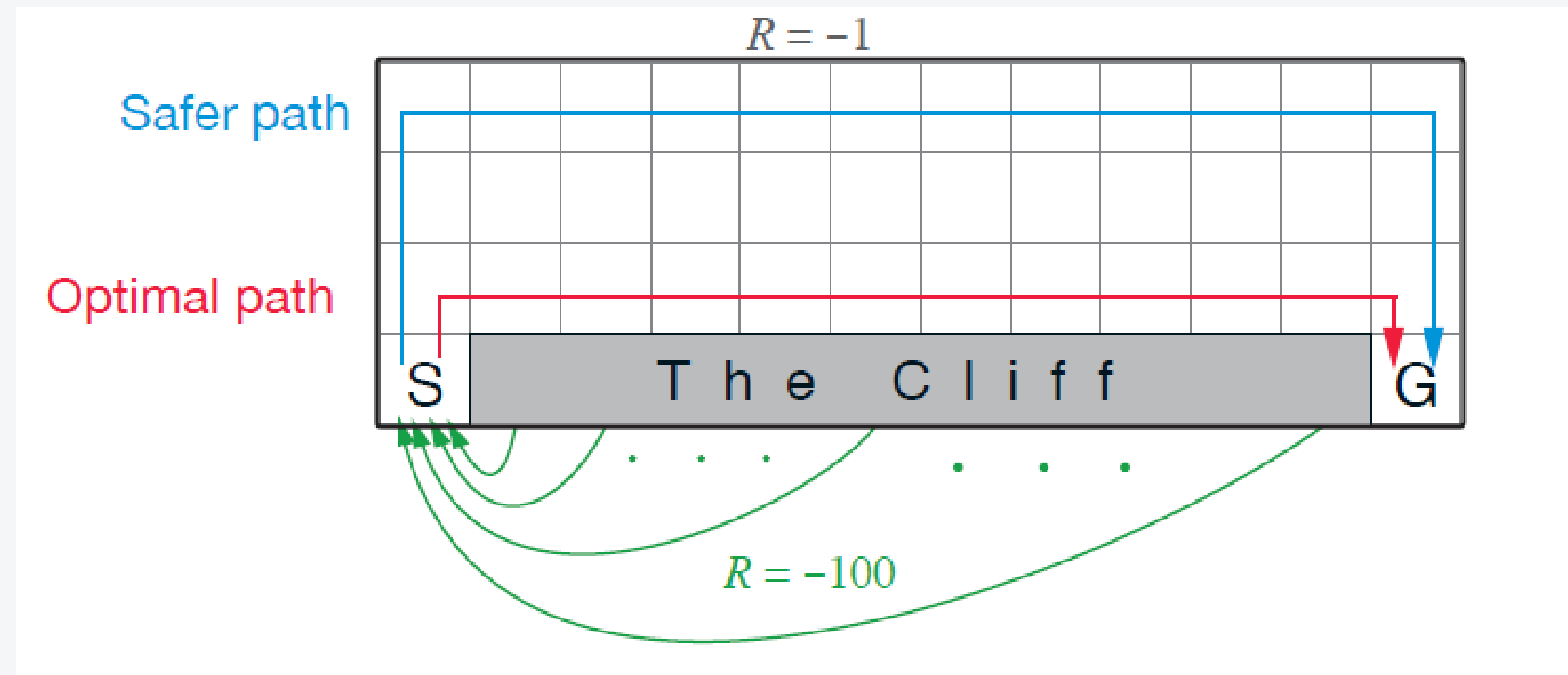
On Policy Learning



On Policy

- So far, we have studied on policy learning.
- We are learning a policy and then corrupting it with epsilon-greedy randomness to do exploration.
- This exploration effects our Q values and leads to a sub-optimal policy.

On Policy

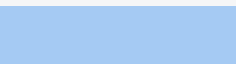


Why SARSA learns the safer path?

$$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma Q(S', A') - Q(S, A)]$$

06.

Off Policy Learning



A Dilemma

- Learning action values conditional on subsequent optimal behavior, but behave non-optimally to explore all actions.
- How can they learn about the optimal policy while behaving according to an exploratory policy?
- The on-policy approach is actually a compromise—it learns action values not for the optimal policy, but for a near-optimal policy that still explores.

Off Policy

- Two separate policies:
- Target policy — what we're learning (becomes the optimal policy)
- Behavior policy — what we act with (explores freely)
- Learning happens from data generated **off the target policy**

Q-Learning

Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$

Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$

Loop for each episode:

 Initialize S

 Loop for each step of episode:

 Choose A from S using policy derived from Q (e.g., ε -greedy)

 Take action A , observe R, S'

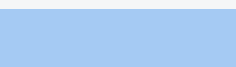
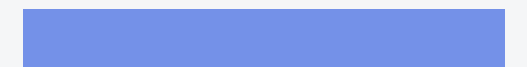
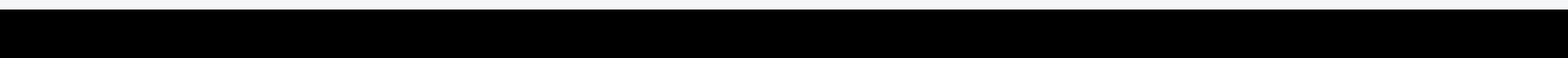
$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_a Q(S', a) - Q(S, A)]$

$S \leftarrow S'$

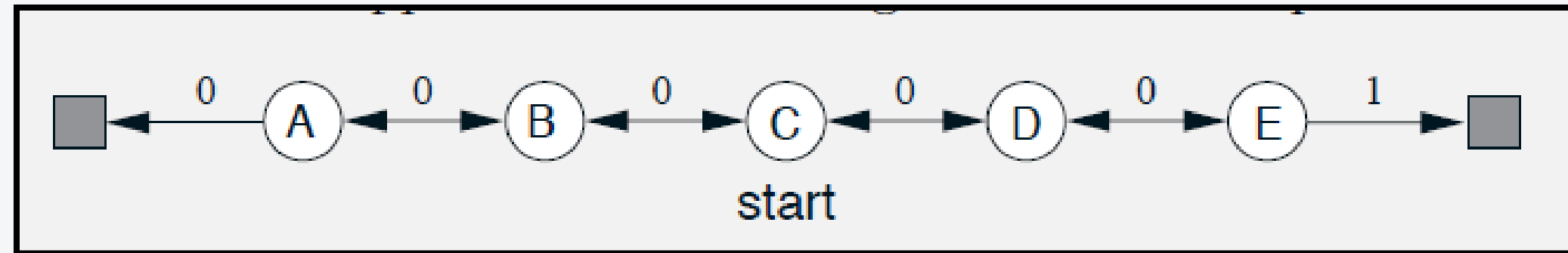
 until S is terminal

07.

Example

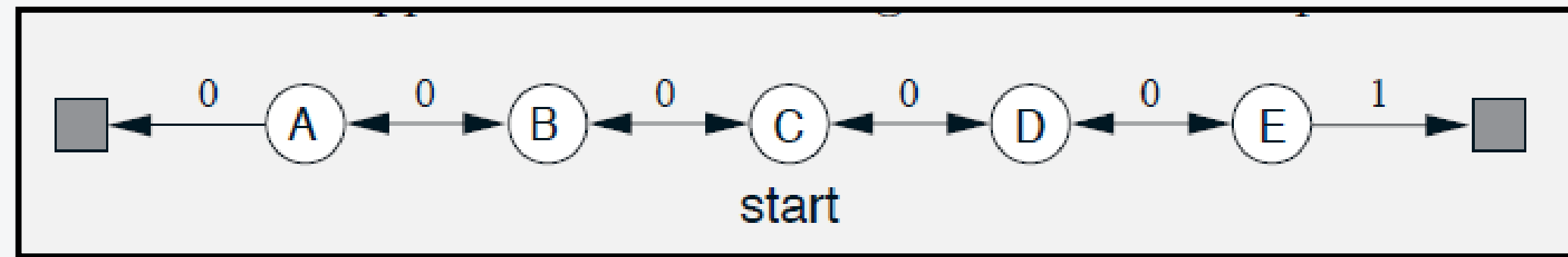


Symmetric Random Walk



Q: What are the true state values?

Symmetric Random Walk



Episode-1: C,0,B,0,C,0,D,0,E,1

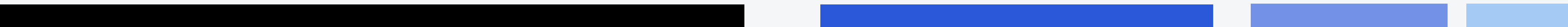
Episode-2: C,0,D,0,C,0,B,0,C,0,D,0,E,1

Episode-3: C,0,B,0,A,0

Q: Apply FVMC, EVMC and TD

08.

Implementation



Frozen Lake

Frozen Lake



Q: Implement SARSA and Q Learning?

[Q-Learning Code](#)

**Thank
You!**

